ICAI AICA Level 2 Capstone Project

Vendor Invoice Portal

Vendor submission and Finance review

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Business need and process

A controlled route for vendor invoices

The portal brings registration and invoice review into one application. Finance controls vendor access and checks the financial fields before export.

Intended benefit

A visible submission reference and review status for each invoice.

01 Vendor registration

02 Finance approves vendor access

03 Invoice and purchase order upload

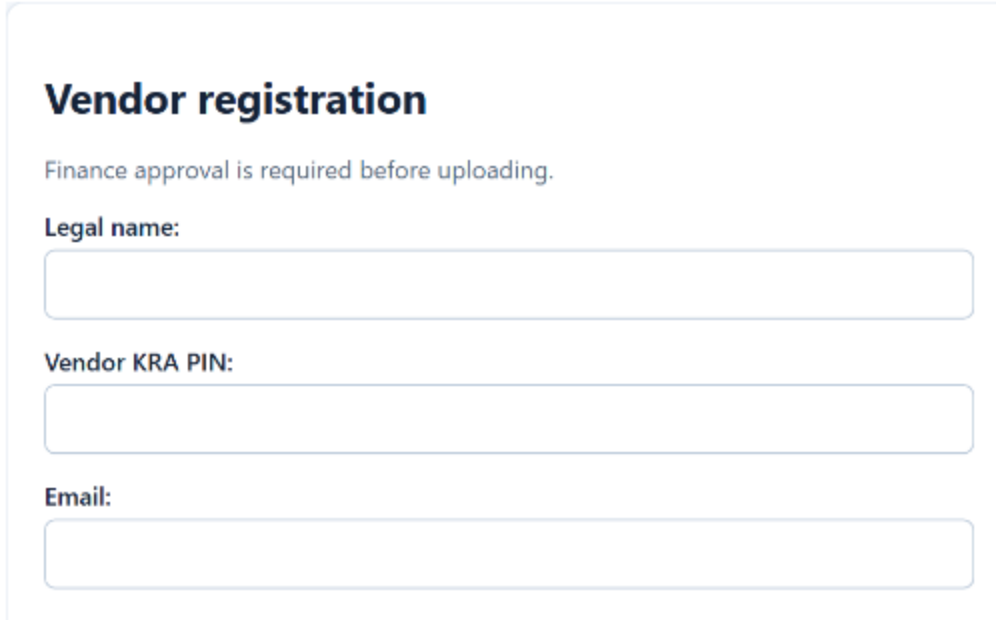
04 Local extraction and validation

05 **Finance verifies and decides**

06 Approved invoices enter CSV export

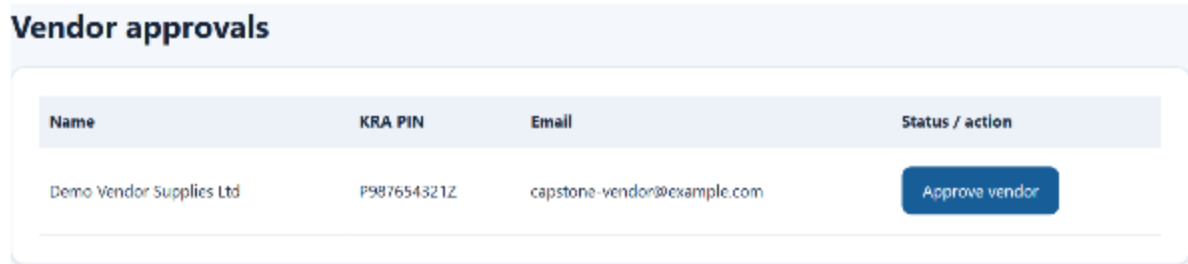
Registration and vendor approval

Vendor registers



Name, Kenyan-format PIN and email

Finance controls access

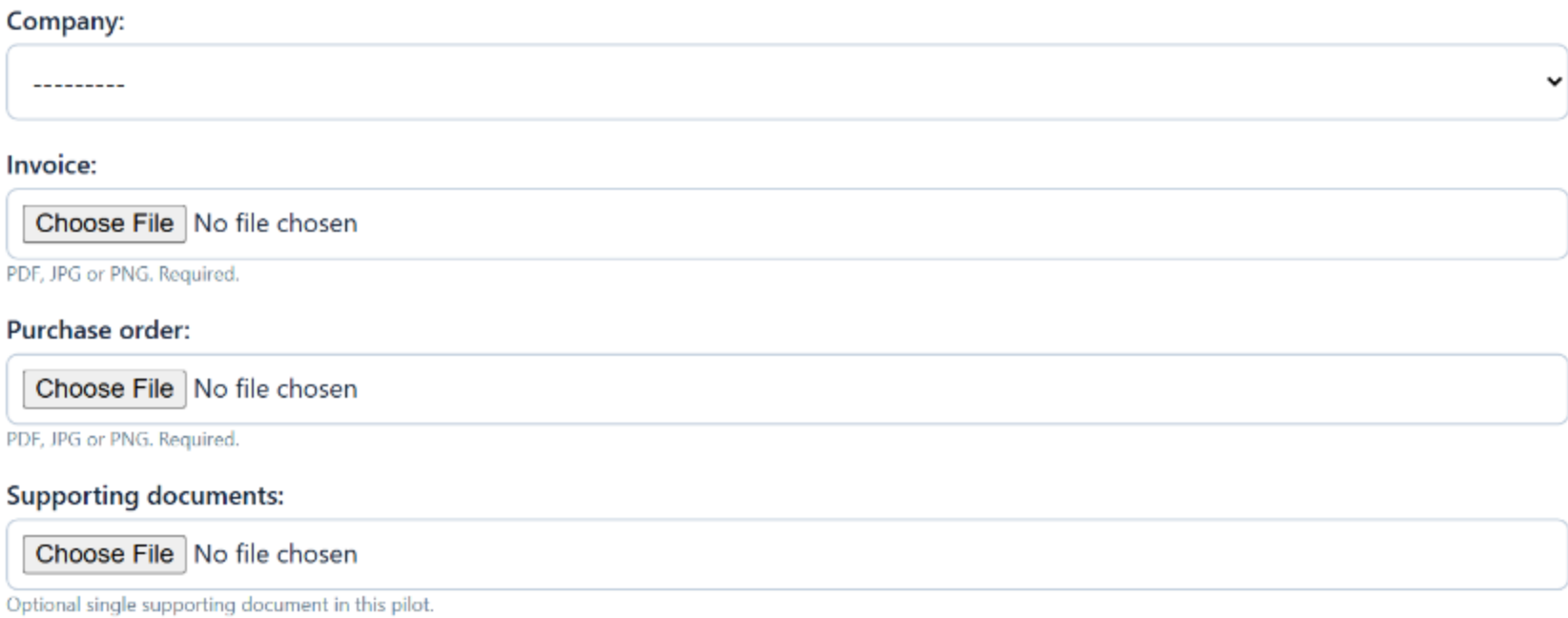


Uploads stay blocked until Finance approves the vendor registration.

Demonstration screens use fictional vendor details

Invoice and purchase order upload

Invoice and PO required



20 MB

combined limit
20,000,000 bytes

PDF, JPG, PNG

Optional supporting document

Recipient company PIN must match the selected company

Validation and approval checks

Vendor approval	Finance must approve the registered vendor before any invoice upload.
Required documents	One invoice and one purchase order are mandatory. One supporting file is optional.
Recipient PIN	The invoice PIN must match the selected company and its recipient / Bill To context.
Duplicate invoice	The same vendor and company cannot reuse an invoice number unless the earlier submission was rejected.
File validation	PDF, JPG or PNG with matching file content. Combined limit: 20,000,000 bytes.
Finance review	Required fields must be complete. Net + VAT must equal total within 0.02 before approval.

Extracted invoice details

Status	**Recipient company**	**Vendor**
Pending finance review	Demo Group Company 01 Ltd	Demo Vendor Supplies Ltd
Invoice number	**Invoice date**	**Currency**
INV-DEMO-0001	Nov. 5, 2026	KES
Net	**VAT**	**Total**
1000.00	160.00	1160.00
PO number		
PO-DEMO-101		

Digital PDF text supplies the initial field values

Finance must verify every value against the attached documents

Finance verification and decision



Verify against originals

Correct invoice fields
and add Finance notes

Net + VAT = Total

Approval tolerance: 0.02

Save, approve or reject

Approved invoices and CSV export

Invoice queue

Reference	Vendor	Company	Currency / Total	Status	Action
CO01-2026-000001	Demo Vendor Supplies Ltd	CO01	KES 1160.00	Approved	Review

Export approved invoices

Only approved invoices not previously exported will be included. Finance must check extracted fields before approval.

Download ERP staging CSV

Only approved, previously unexported invoices enter the staging CSV

Architecture and AI contribution

Browser	Vendor and Finance screens
Waitress and Django	Routing, access checks and workflow
Local processing	PDF text, optional Tesseract OCR and parsing rules
Demo storage	SQLite records and protected document downloads

Runtime: local extraction and parsing rules

AI assisted development and submission preparation

Verification and prototype limits

7 automated checks passed

Verified paths

Registration and approval gating
Invoice validation and extraction
Finance review and CSV export
Vendor isolation and CSRF checks

Further validation

Exact Windows Python 3.13 setup
Scanned document OCR
Concurrent uploads and exports
Production security and recovery

Test environment: Python 3.12 with Pillow 13.0.0

The exact pinned environment and public deployment remain unverified

Capstone submission

Sanjeev Panwar

GitHub: Sanjeev-Panwar899

One clean project folder

Source code, README, dependencies, setup instructions and project notes

Fork and pull request

Target repository: aiinicai/AICA-Level-2-Projects

Fictional data only. Private files and credentials stay out of GitHub.

This presentation is optional under the supplied repository upload guide